# Root Cause Analysis Report

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|---|---|
| Part | MS-4505-001 Sheet Metal Bracket |
| Material / process | SPCC Steel / bending |
| Case | rca_fcada75f04f14f4ea9232d597d30e7b5 |
| Date | 2026-11-01 17:57:06 |
| Status | AI proposal, pending expert review |

## 1. Problem

| Defect | Severity | Confidence | Detail |
|---|---|---|---|
| size_mismatch | major | 99% | length 118.44 mm (nom 120.00 ±0.10) |
| size_mismatch | major | 99% | hole dia 12.73 (nom 12.00 ±0.05) |
| size_mismatch | major | 99% | hole position off 1.91 mm (±0.10) |

## 2. Root cause

**Cause:** Improper Bending Force / Over Bending

**Reason:** Reason: Major size mismatch defects during press brake bending operations align with improper force settings beyond the material elastic limit, which leads to dimensional inaccuracy and deformation in bent features [586f086d-0deb-40c2-a71d-f840aaa55a60].

**Effect:** Deformation, Dimensional Inaccuracy

**Confidence:** 95%

**Supporting evidence**
- **586f086d-0deb-40c2-a71d-f840aaa55a60**: Improper Bending Force / Over Bending Root Cause Analysis Defect Types: deformation, dimensional inaccuracy Process: sheet metal bending, press brake operations Root Cause: Improper bending force settings or over-bending beyond the material's elastic limit. Effect: Deformation of the part, dimensional inaccuracy in bent features, springback issues causing part to not meet dimensional tolerances. C

## 3. Solution

### *Corrective actions (fix this occurrence)*

1. Recalibrate press brake force settings to material specification
2. Verify tonnage calculator against actual material hardness test results
3. Implement first-article inspection

### *Preventive actions (stop it recurring)*

1. Establish machine force calibration schedules and validated tonnage charts

### *Precautions*

- Ensure press brake operator verifies SPCC steel thickness and hardness before set up

### *How to verify the fix*

Re-check part dimensions after press brake recalibration

## 4. Other causes considered

| Rank | Cause | Confidence | Reason |
|---|---|---|---|

| | | | |
|---|---|---|---|
| 2 | Material Hardness Variation | 94% | Reason: Hardness variation in incoming SPCC steel lot causes unpredictable springback behavior during bending operations, resulting in dimensional inaccuracy and size mismatches [02cf0928-6739-4f7e-ae1b-f6740283f00f]. |
| 3 | Springback Elastic Recovery | 0% | Reason: Detected size mismatch defects during sheet metal bending indicate dimensional inaccuracy in bent features. As detailed in [6872e1ff-9d0f-41f8-9cd9-c783d66c1b46], elastic recovery of sheet metal after forming causes the bent angle to open up beyond the intended angle, producing dimensional i |

## 5. Traceability

AI model: gemini:gemini-3.5-flash. Expert edited the cause: no.
*This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.*